
HOMEWORK DAY 7 - Differentiation formulas §2.3

Differentiate the following functions. Simplify your answer.

1. §2.3: 1 $g(x) = 4x + 7$
 $g'(x) = 4$

2. §2.3: 3 $f(x) = x^{15} - x^{13}$
 $f'(x) = 15x^{14} - 13x^{12}$

3. §2.3: 4 $g(x) = \frac{7}{4}x^2 - 3x + 2$
 $g'(x) = \frac{7}{2}x - 3$

4. §2.3: 6 $r(z) = z^{-5} - z^{1/2}$
 $r'(z) = -5z^{-6} - \frac{1}{2}z^{-1/2}$
 $r'(z) = -\frac{5}{z^6} - \frac{1}{2\sqrt{z}}$

5. §2.3: 7 $f(x) = x^{3/2} + x^{-3}$
 $f'(x) = \frac{3}{2}x^{1/2} - 3x^{-4}$
 $f'(x) = \frac{3}{2}\sqrt{x} - \frac{3}{x^4}$

6. §2.3: 9 Answer: $s(t) = t^{-1} + t^{-2}$, $s'(t) = -t^{-2} - 2t^{-3} = -\frac{1}{t^2} - \frac{2}{t^3} = -\frac{t+2}{t^3}$ (last step is optional)

7. §2.3: 10 $r(t) = \frac{a}{t^2} + \frac{b}{t^4}$
 $= at^{-2} + bt^{-4}$
 $r'(t) = -2at^{-3} - 4bt^{-5}$
 $r'(t) = -\frac{2a}{t^3} - \frac{4b}{t^5}$

8. §2.3: 11 $y = 2x + \sqrt{x}$
 $y = 2x + x^{1/2}$
 $y' = 2 + \frac{1}{2}x^{-1/2}$
 $y' = 2 + \frac{1}{2\sqrt{x}}$

9. §2.3: 12 $h(w) = \sqrt{2}w - \sqrt{2}$
 $h'(w) = \sqrt{2} - 0$
 $h'(w) = \sqrt{2}$

10. §2.3: 13 $g(x) = \frac{1}{\sqrt{x}} + 4\sqrt{x}$
 $g(x) = x^{-1/2} + x^{1/2}$
 $g'(x) = -\frac{1}{2}x^{-3/2} + \frac{1}{2}x^{-1/2}$ $\nearrow$ $-\frac{1}{2\sqrt{x^3}} + \frac{1}{4\sqrt{x}}$

11. §2.3: 34 $v = \frac{y^2}{1-x}$ $v' = \frac{(1-x)2y - y^2(-1)}{(1-x)^2}$ $\nearrow$ $v' = \frac{2x - y^2}{1-x^2}$
 $v' = \frac{2x - 2y^2 + y^2}{(1-x)^2}$

12. §2.3: 35 $g(t) = \frac{3-2t}{5t+1}$ $g'(t) = \frac{5t+1(-2) - 3-2t(5)}{(5t+1)^2}$ $\nearrow$ $g'(t) = \frac{-17}{(5t+1)^2}$
 $= \frac{-10t-2-15-10t}{(5t+1)^2}$

13. §2.3: 39 $y = \frac{s-\sqrt{s}}{s^2}$ $y = \frac{s}{s^2} - \frac{\sqrt{s}}{s^2}$ $\nearrow$ $y' = -s^{-2} + \frac{3}{2}s^{-5/2}$
 $= \frac{1}{s} - \frac{s^{1/2}}{s^2}$
 $= \frac{1}{s} - \frac{1}{s^{3/2}}$
 $y = s^{-1} - s^{-3/2}$
 $= -\frac{1}{s^2} + \frac{3}{2s^{5/2}}$
 $= -\frac{1}{s^2} + \frac{3}{2\sqrt{s^5}}$

14. §2.3: 43 $h(u) = (u^2 - u)(u + 1)$ $\nearrow$ $h'(u) = 2u - 1$
 $h(u) = (u^2 - u)$

15. §2.3: 46 $h(w) = (w^2 + 3w)(w^{-1} - w^{-4})$
 $h(w) = w - w^2 + 3 - 3w^3$
 $h'(w) = 1 - 2w + 0 - 9w^2$
 $h'(w) = -9w^2 - 2w + 1$

16. §2.3: 27

$$f(x) = (1 + 2x^2)(x - x^2)$$

Product rule: $\frac{d}{dx}(f(x)g(x)) = f(x)g'(x) + g(x)f'(x)$

$$f'(x) = 1 + 2x^2(1 - 2x) + x - x^2(4x)$$

$$= 1 - 2x + 2x^2 - 4x^3 + x - 4x^3$$

$$f'(x) = -2x^3 + 6x^2 - 2x + 1$$

Answers correct

$$P(x) = (1 + 2x^2)(x - x^2)$$

$$P(x) = x - x^2 + 2x^3 - 2x^4$$

$$P'(x) = 1 - 2x + 6x^2 - 8x^3$$

$$P'(x) = -8x^3 + 6x^2 - 2x + 1$$

17. §2.3: 28

$$F(x) = \frac{x^4 - 5x^3 + \sqrt{x}}{x^2}$$

$$F(x) = \frac{x^4}{x^2} - \frac{5x^3}{x^2} + \frac{x^{1/2}}{x^2}$$

$$= x^2 - 5x + \frac{1}{x^{3/2}}$$

$$= x^2 - 5x + x^{-3/2}$$

$$F'(x) = 2x - 5 - \frac{3}{2}x^{-5/2}$$

$$F'(x) = 2x - 5 - \frac{3}{2\sqrt{x^5}}$$

$$P'(x) = \frac{x^2(4x^3 - 15x^2 + \frac{1}{2}x^{-1/2}) - (x^4 - 5x^3 + \sqrt{x})(2x)}{(x^2)^2}$$

$$= \frac{4x^5 - 15x^4 + \frac{1}{2}x^{3/2} - (2x^5 - 10x^4 + 2x^{3/2})}{x^4}$$

$$= \frac{2x^5 - 5x^4 - \frac{3}{2}x^{3/2}}{x^4}$$

$$= \frac{2x^5}{x^4} - \frac{5x^4}{x^4} - \frac{3x^{3/2}}{2x^4}$$

$$= 2x - 5 - \frac{3}{2}x^{-5/2}$$

$$F'(x) = 2x - 5 - \frac{3}{2\sqrt{x^5}}$$

≠ like the simplifying for

bc the expressions stay smaller and there isn't fooling/simplifying on a large scale

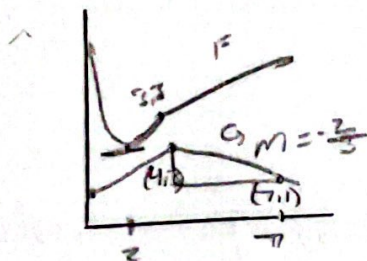
18. §2.3: 79(a,b) $f(s)=1, f'(s)=6$
 $g(s)=-3, g'(s)=2$

A. $(fg)'(s) = fg' + gf'$
 $= 1(2) + (-3)(6)$
 $= 2 - 18$
 $fg' = -16$

B. $(f/g)'(s) = \frac{gf' - fg'}{g^2}$
 $= \frac{-3(6) - 1(2)}{-3^2}$
 $= \frac{-18-2}{9}$

$(\frac{f}{g})'(s) = -\frac{20}{9}$

19. §2.3: 84 $P(x) = F(x)G(x)$
 $Q(x) = F(x)/G(x)$



A. find $P'(2)$

$P'(2) = 0$

B. find $Q'(7)$

$Q'(7) = -\frac{2}{3}$

20. §2.3: 86(a,b)

a) $y = x^2 f(x)$

$y' = x^2 f'(x) + f(x) 2x$

b) $y = \frac{f(x)}{x^2}$

$y' = \frac{x^2 f'(x) - f(x) 2x}{x^4}$

$$\sin = y \quad \tan = \frac{y}{x}$$

$$\csc = \frac{1}{\sin} \quad \cot = \frac{\cos}{\sin}$$

$$\sec = \frac{1}{\cos}$$

HOMEWORK DAY 8 - Derivatives of Trigonometric functions §2.4

Find the following derivatives. Simplify if appropriate.

21. §2.4: 2. $f(x) = \tan x - 4 \sin x$

$$f'(x) = \sec^2 x - 4 \cos x$$

$$f'(x) = \frac{1}{\cos^2 x} - 4 \cos x$$

$$f'(x) = \frac{1 - 4 \cos^3 x}{\cos^2 x}$$

22. §2.4: 3. $y = x^2 + \cot x$

$$y = x^2 + \csc^2 x$$

23. §2.4: 5. $h(\theta) = \theta^2 \sin \theta$

$$h'(\theta) = \theta^2 \cos \theta + 2\theta \sin \theta$$

$$h(\theta) = \theta(\theta \cos \theta + 2 \sin \theta)$$

24. §2.4: 6. $g(x) = 3x + x^2 \cos x$

$$g'(x) = 3 + x^2(-\sin x) + 2x \cos x$$

$$g'(x) = 3 - x^2 \sin x + 2x \cos x$$

25. §2.4: 7. Answer: $\frac{dy}{d\theta} = \sec \theta \tan \theta \tan \theta + \sec \theta \sec^2 \theta = \sec \theta (\tan^2 \theta + \sec^2 \theta)$

26. §2.4: 8. $y = \sin \theta \cos \theta$

$$y' = \sin \theta (-\sin \theta) + \cos \theta (\cos \theta)$$

$$= -\sin^2 \theta + \cos^2 \theta$$

$$v' = \cos(2\theta)$$

27. §2.4: 10.

$$f(x) = x \cos x + 2 \tan x$$

$$f'(x) = \cos x + x(-\sin x) + 2 \sec^2 x + \tan x(0)$$

$$f'(x) = \cos x - x \sin x + 2 \sec^2 x$$

28. §2.4: 40 (A point on a curve has two coordinates (a, b))

$$y = \frac{\cos x}{2 + \sin x}$$

$$y' = \frac{2 + \sin x (-\sin x) - [\cos x (\cos x)]}{(2 + \sin x)^2}$$

$$= \frac{-2\sin x - \sin^2 x - (\cos^2 x)}{(2 + \sin x)^2}$$

$$-\sin^2 x - \cos^2 x = -1$$

$$= \frac{-2\sin x - \sin^2 x - \cos^2 x}{(2 + \sin x)^2}$$

$$= \frac{-2\sin x - 1}{(2 + \sin x)^2}$$

$$y' = -\frac{2\sin x + 1}{(2 + \sin x)^2}$$

Horizontal tangent

② $y' = 0$, only numerator

has to be 0 $\in \mathbb{D}: -\infty, \infty$



$$2\sin x + 1 = 0$$

$$\sin x = -\frac{1}{2}$$

$$x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\frac{\cos(-\frac{\pi}{6})}{2 + \sin(-\frac{\pi}{6})} = \frac{\frac{\sqrt{3}}{2}}{2 - \frac{1}{2}} = \frac{\sqrt{3}}{4-1} = \boxed{\frac{\sqrt{3}}{3}}$$

$$\frac{\cos(\frac{7\pi}{6})}{2 + \sin(\frac{7\pi}{6})} = \frac{-\frac{\sqrt{3}}{2}}{2 - \frac{1}{2}} = \frac{-\sqrt{3}}{4-1} = \boxed{-\frac{\sqrt{3}}{3}}$$

$$\left\{ \begin{aligned} & \left(-\frac{\pi}{6} + 2\pi k, \frac{\sqrt{3}}{3} \right) \\ & \left(\frac{7\pi}{6} + 2\pi k, -\frac{\sqrt{3}}{3} \right) \end{aligned} \right\} \quad \begin{aligned} & k \text{ is an integer} \\ & k \in \mathbb{Z} \end{aligned}$$